

Compressed Sensing with Quantized Tensor Trains (QTTs)

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Abstract

The storage and recovery of a vector of length 2^d can become prohibitively expensive as d grows, a manifestation of the curse of dimensionality. For certain structured functions and multiscale quantities, the resulting discretized vectors admit compact quantized tensor train (QTT) representations after binary tensorization: a length- 2^d vector is reshaped into a d -way binary tensor with low tensor-train (TT) rank. Motivated by this construction, we study compressed sensing of tensorized signals that combine bounded TT rank with sparse low-order interaction structure, using randomly sampled Walsh–Hadamard measurements. For this structured model class, we establish a uniform restricted isometry property and derive a measurement condition guaranteeing uniqueness of noiseless recovery. We also propose a structure-aware initializer for the alternating linear scheme (ALS), obtained by projecting the adjoint backprojection onto the low-order interaction space before applying TT-SVD. We prove a uniform initialization error bound whose measurement requirement, for fixed structural parameters, grows polynomially with the tensor order d rather than with the ambient dimension 2^d . Numerical experiments support the predicted RIP and initialization scaling and demonstrate more reliable ALS recovery.

Keywords: quantized tensor trains, compressed sensing, Walsh–Hadamard measurements, restricted isometry property, low-order interactions, alternating linear schemes

1 Introduction

Recovering a high-dimensional signal from a limited number of measurements is a central problem in inverse problems, signal processing, and scientific computing. Without additional structural assumptions, unique recovery of a general vector in $\mathbb{R}^N$ requires at least N linear measurements. Classical compressed sensing reduces this burden by assuming coordinate sparsity. If $\mathbf{x} \in \mathbb{R}^N$ has at most q nonzero entries, one observes

$$\mathbf{y} = \mathbf{A}\mathbf{x} + \boldsymbol{\eta},$$

where $\mathbf{A} \in \mathbb{R}^{m \times N}$, $m \ll N$, and $\boldsymbol{\eta} \in \mathbb{R}^m$ represents measurement noise or outliers. When $\mathbf{A}$ satisfies a restricted isometry property (RIP), sparse vectors can be recovered by ℓ_1 minimization or greedy methods [4, 3, 8]. Dense Gaussian and Bernoulli matrices can achieve near-optimal measurement rates, but storing and applying them becomes costly in large ambient dimensions. Subsampled Fourier matrices are an important structured alternative: they support fast matrix–vector multiplication through the fast Fourier transform and arise naturally in frequency-domain sensing, notably magnetic resonance imaging (MRI) [16]. In the binary setting considered here, the Walsh–Hadamard transform provides a natural alternative:

its recursive Kronecker structure likewise permits a butterfly implementation with $\mathcal{O}(N \log N)$ complexity and aligns naturally with binary tensorization.

Let $N = 2^d$ and let $\mathbf{x}^* \in \mathbb{R}^N$ be the unknown vector. Its binary tensorization

$$\mathcal{X}^* = \text{ten}(\mathbf{x}^*) \in \mathbb{R}^{2 \times \cdots \times 2}$$

is defined precisely in Section 2.1. A tensor-train (TT) representation of this tensorized vector is called a quantized tensor train (QTT).¹ We assume that $\mathcal{X}^*$ has bounded TT rank and belongs to a sparse low-order interaction model, as formalized in Section 2.4. Each binary mode represents one bit of the original vector index. For function samples on a grid with 2^d points, these bits can be associated with scales ranging from coarse to fine. Low TT rank then models limited dependence across these levels [29]. If the ranks are bounded by r , the QTT storage is $\mathcal{O}(dr^2)$ rather than $\mathcal{O}(2^d)$; see Section 2.2.

Complementing the low-TT-rank prior, the sparse low-order interaction model assumes that $\mathcal{X}^*$ has a sparse expansion in components involving only a few binary modes. This assumption is motivated by physical models dominated by a small number of low-order interactions [25, 6]. For a generic tensorized vector, it represents structured sparsity among the binary index variables. A closely related combination of sparse low-order interaction structure and TT compressibility is used for density estimation by Peng et al. [21], while a related low-order cluster basis and sparse-plus-low-rank compression are used for operator learning by Khoo et al. [14]. Together, the two assumptions combine global QTT compressibility with sparse low-order interaction structure.

We acquire m randomly selected Walsh–Hadamard measurements. Let

$$\mathbf{H}_2 := \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}, \quad \mathbf{H} := \mathbf{H}_2^{\otimes d} \in \mathbb{R}^{N \times N},$$

and let $\mathbf{S} \in \mathbb{R}^{m \times N}$ select coordinates independently and uniformly with replacement. With the normalization used throughout the analysis, the noiseless measurements are

$$\mathbf{y} = \mathbf{A} \mathbf{x}^*, \quad \mathbf{A} := \sqrt{\frac{N}{m}} \mathbf{S} \mathbf{H}. \quad (1)$$

Because $\mathbf{H}_2$ is the normalized two-point discrete Fourier transform, this measurement model can equivalently be viewed as random sampling of the d -dimensional discrete Fourier transform of $\mathcal{X}^*$ on a $2 \times \cdots \times 2$ grid. Given $\mathbf{y}$ and the sampled rows, our goal is to recover $\mathbf{x}^*$, or equivalently its tensorization $\mathcal{X}^*$.

The present work is closely related to two lines of research: low-rank tensor recovery and tensor regression.

Tensor recovery. Through binary tensorization, the recovery problem above can be viewed as low-rank tensor recovery from general linear measurements. Rauhut, Schneider, and Stojanac [26] introduced iterative hard-thresholding algorithms for recovering low-rank tensors in several formats and analyzed these algorithms under a tensor restricted isometry property (TRIP). Their bounds also cover partial Fourier maps combined with random sign flips. For tensor recovery and other optimization problems in the TT format, Holtz, Rohwedder, and Schneider [11] developed the alternating linear scheme (ALS), which is the recovery algorithm used in this work. An important special case of low-rank tensor recovery is tensor completion, where the measurement operator observes a subset of tensor entries. Steinlechner [30]

¹Here, “quantized” refers to binary tensorization and is distinct from scalar quantization of measurements; see Hou and Ng [12] for tensor recovery from quantized modewise measurements.

formulated fixed-TT-rank tensor completion as Riemannian optimization, and Cai, Li, and Xia [2] subsequently developed a provable second-order spectral initialization and established linear convergence under a nearly optimal sampling condition.

More recently, for general linear measurements of low-TT-rank tensors, Qin, Wakin, and Zhu [23] established local linear convergence of factorized Riemannian gradient descent under TT-RIP, using the spectral initializer $\text{TT-SVD}_r(\mathcal{A}^*\mathbf{y})$. Qin and Zhu [24] later treated arbitrary outliers using an ℓ_1 loss and Riemannian subgradient methods. Our initializer uses the same adjoint backprojection followed by TT-SVD, but inserts a projection onto the order- s interaction space between them, giving $\text{TT-SVD}_r(\mathcal{P}_{\mathcal{V}_s}(\mathcal{A}^*\mathbf{y}))$. In our experiments, the added interaction-space projection yields lower initialization error and more reliable ALS recovery than the unprojected spectral initializer; it also underlies the uniform initialization guarantee developed below.

Tensor regression. Using the tensor notation introduced in Section 2.1, let $f := \mathcal{H}(\mathcal{X}^*)$. As explained in Section 2.4, f admits a sparse expansion in the Walsh tensor-product basis, truncated to interactions of order at most s . Moreover,

$$\mathbf{y} = \sqrt{\frac{N}{m}} \mathbf{S} \text{vec}(f),$$

so the observations are random point samples of f . This interpretation connects our problem to sparse low-rank approximation of multivariate functions from point evaluations. Chevreuil et al. [5] construct sparse canonical low-rank tensor-product approximations from random function evaluations using regularized least squares and greedy rank-one enrichment. In related work of analysis of variance (ANOVA) decompositions, sparse approximation from scattered samples has been studied using Fourier-based least-squares and regularization methods [22, 1].

The coefficient tensor of a tensor-product expansion can itself be represented and compressed in tensor-network formats; see, e.g., [33]. Particularly relevant is the block-sparse TT regression framework of Götte, Schneider, and Trunschke [9]. They represent the coefficients of a multivariate polynomial tensor-product expansion in TT format and fit the model from point evaluations using an ALS-type least-squares method. Thus, both settings infer a TT-structured coefficient tensor from pointwise observations of the corresponding tensor-product function. The structural assumptions are different, however: Götte et al. use a polynomial tensor-product basis with block sparsity induced by polynomial degree, whereas we use the Walsh/ANOVA cluster basis and further assume sparsity among the active low-order interaction components. Their analysis focuses on the resulting block-sparse TT representation and its complexity, while we establish uniform RIP and noiseless uniqueness guarantees for subsampled Walsh–Hadamard measurements and develop a structure-aware initializer for ALS.

A closely related contemporaneous work is presented in Chapter 6 of Peng’s dissertation [20, Chapter 6]. Both works represent sparse low-order interactions in an ANOVA tensor-product basis and combine this structure with a TT representation, so their structural starting points are closely aligned. At the level of the functional representation and point-sampling observation model, our setting is the binary two-basis specialization of Peng’s framework. The methods and theoretical guarantees, however, are different. Peng combines kernel ridge regression, TT compression, coefficient-based feature sampling, and a reduced sparse-regression refit. In contrast, we study direct recovery under a randomly subsampled Walsh–Hadamard operator, establish a uniform RIP and noiseless uniqueness guarantee for the joint structured model, and develop a structure-aware initializer for fixed-rank ALS together with a uniform initialization error bound.

1.1 Contributions

The contributions of this work are summarized as follows.

- This work provides a compressed-sensing framework for recovering vectors whose binary tensorizations combine bounded TT rank with sparse low-order interaction structure.
- A uniform RIP is established for the randomly sampled Walsh–Hadamard operator on the resulting model class and is later used to prove uniqueness of noiseless recovery.
- A structure-aware initializer for ALS is proposed and shown to satisfy a uniform initialization error bound; for fixed structural parameters, the required number of measurements grows polynomially in d .
- Numerical experiments examine the predicted RIP and initialization scalings and compare the structure-aware initializer with alternative initializations in end-to-end ALS recovery.

1.2 Organization

The remainder of this paper is organized as follows. Section 2 introduces the notation, preliminaries, and problem setup. Section 3 presents the ALS recovery method and the proposed initialization. Section 4 establishes the structured RIP and uniqueness of noiseless recovery. Section 5 analyzes the structure-aware initialization. Section 6 reports the numerical experiments, and Section 7 concludes the paper.

2 Preliminaries

2.1 Notation

To distinguish the tensor formulation from its vectorized representation, we use the typographical conventions summarized below.

Notation type	Objects	Examples
Lowercase italic	Scalars	d, m, N, r
Bold lowercase	Vectors	$\mathbf{x}, \mathbf{y}$
Bold uppercase	Matrices	$\mathbf{A}, \mathbf{H}$
Calligraphic uppercase	Tensors and tensor operators	$\mathcal{X}, \mathcal{A}, \mathcal{H}$
Fraktur uppercase	Model classes	$\mathfrak{M}_{r,s,k}(C)$

We use $\mathbb{R}$ and $\mathbb{C}$ to denote the real and complex number fields, respectively. For a positive integer n , we write $[n] := \{1, \dots, n\}$. We use $\otimes$ for the Kronecker product and $\circ$ for the tensor outer product of vectors; their relation under our vectorization convention is specified below.

Tensors and vectorization. The tensor order is denoted by d . Since every mode has size two, the ambient vector dimension is $N := 2^d$. For $\mathcal{X} \in \mathbb{R}^{2 \times \dots \times 2}$, we write

$$\mathbf{x} := \text{vec}(\mathcal{X}) \in \mathbb{R}^N \quad (2)$$

for its vectorization. More precisely, for $i_1, \dots, i_d \in \{1, 2\}$, we define

$$[\text{vec}(\mathcal{X})]_j = \mathcal{X}(i_1, \dots, i_d), \quad j = 1 + \sum_{k=1}^d (i_k - 1)2^{k-1}. \quad (3)$$

Thus, the first tensor index varies fastest. We denote the inverse tensorization map by

$$\text{ten} := \text{vec}^{-1}, \quad \text{ten}(\text{vec}(\mathcal{X})) = \mathcal{X}. \quad (4)$$

For matrices $\mathbf{A} = [a_{ij}] \in \mathbb{R}^{m \times n}$ and $\mathbf{B} \in \mathbb{R}^{p \times q}$, their Kronecker product is the block matrix

$$\mathbf{A} \otimes \mathbf{B} := [a_{ij}\mathbf{B}]_{i=1,\dots,m; j=1,\dots,n} \in \mathbb{R}^{mp \times nq}. \quad (5)$$

For vectors $\mathbf{a}^{(k)} \in \mathbb{R}^{n_k}$, $k = 1, \dots, d$, their tensor outer product is the d -way rank-one tensor

$$\mathcal{Z} := \mathbf{a}^{(1)} \circ \dots \circ \mathbf{a}^{(d)} \in \mathbb{R}^{n_1 \times \dots \times n_d}, \quad \mathcal{Z}(i_1, \dots, i_d) = \prod_{k=1}^d a_{i_k}^{(k)}. \quad (6)$$

Under the vectorization convention above,

$$\text{vec}(\mathcal{Z}) = \mathbf{a}^{(d)} \otimes \dots \otimes \mathbf{a}^{(1)}. \quad (7)$$

Thus, $\circ$ constructs a multiway tensor, whereas $\otimes$ denotes the corresponding product in vectorized coordinates. The reversed order in the last display follows because the first tensor index varies fastest.

For vectors $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$, define the Euclidean inner product by $\langle \mathbf{u}, \mathbf{v} \rangle_2 := \mathbf{u}^\top \mathbf{v}$. For $\mathcal{X}, \mathcal{Y} \in \mathbb{R}^{2 \times \dots \times 2}$, the Frobenius inner product is defined as

$$\langle \mathcal{X}, \mathcal{Y} \rangle_F := \sum_{i_1=1}^2 \dots \sum_{i_d=1}^2 \mathcal{X}(i_1, \dots, i_d) \mathcal{Y}(i_1, \dots, i_d). \quad (8)$$

The corresponding Frobenius norm satisfies

$$\|\mathcal{X}\|_F = \sqrt{\langle \mathcal{X}, \mathcal{X} \rangle_F} = \|\text{vec}(\mathcal{X})\|_2. \quad (9)$$

For vectors and matrices, $\|\cdot\|_2$ denotes the Euclidean norm and the spectral norm, respectively.

Using these inner products, the adjoint $\mathcal{L}^* : \mathbb{R}^m \rightarrow \mathbb{R}^{2 \times \dots \times 2}$ of a linear operator $\mathcal{L} : \mathbb{R}^{2 \times \dots \times 2} \rightarrow \mathbb{R}^m$ is defined by

$$\langle \mathcal{L}(\mathcal{X}), \mathbf{y} \rangle_2 = \langle \mathcal{X}, \mathcal{L}^*(\mathbf{y}) \rangle_F \quad (10)$$

for all $\mathcal{X}$ and $\mathbf{y}$. For a real matrix $\mathbf{L}$, the adjoint is its transpose: $\mathbf{L}^* = \mathbf{L}^\top$.

Walsh–Hadamard transform. Recall the normalized Walsh–Hadamard matrix $\mathbf{H}$ introduced above. We denote its corresponding modewise tensor transform by $\mathcal{H}$, where

$$\text{vec}(\mathcal{H}(\mathcal{X})) = \mathbf{H} \text{vec}(\mathcal{X}). \quad (11)$$

Every entry of $\mathbf{H}$ has magnitude $2^{-d/2} = N^{-1/2}$. Moreover, $\mathbf{H}$ is real, symmetric, and orthogonal, so

$$\mathbf{H}^{-1} = \mathbf{H}^* = \mathbf{H}, \quad \mathcal{H}^{-1} = \mathcal{H}^* = \mathcal{H}. \quad (12)$$

2.2 Tensor-train decomposition and TT-SVD

We first recall the TT representation and then describe its sequential construction by TT-SVD [18]. The discussion allows general mode sizes, although this paper focuses on $n_1 = \dots = n_d = 2$.

Tensor-train representation. Let $\mathcal{X} \in \mathbb{R}^{n_1 \times \dots \times n_d}$. A TT representation consists of cores $\mathcal{G}_k \in \mathbb{R}^{r_{k-1} \times n_k \times r_k}$, $k = 1, \dots, d$, with boundary ranks $r_0 = r_d = 1$. For $i_k = 1, \dots, n_k$, write $\mathbf{G}_k(i_k) := \mathcal{G}_k(:, i_k, :) \in \mathbb{R}^{r_{k-1} \times r_k}$. Then

$$\begin{aligned} \mathcal{X}(i_1, \dots, i_d) &= \mathbf{G}_1(i_1) \mathbf{G}_2(i_2) \cdots \mathbf{G}_d(i_d) \\ &= \sum_{\alpha_1=1}^{r_1} \cdots \sum_{\alpha_{d-1}=1}^{r_{d-1}} \mathcal{G}_1(1, i_1, \alpha_1) \mathcal{G}_2(\alpha_1, i_2, \alpha_2) \cdots \mathcal{G}_d(\alpha_{d-1}, i_d, 1). \end{aligned}$$

The matrix product is a scalar because $r_0 = r_d = 1$; we write the represented tensor compactly as $\mathcal{X} = [\mathcal{G}_1, \dots, \mathcal{G}_d]$. In quantum many-body physics, the TT format is commonly referred to as a matrix product state (MPS) representation [28]. When the tensor is obtained by tensorizing a vector, its TT representation is called a QTT representation, also known as a quantics tensor train [17].

For $k = 1, \dots, d-1$, let $\mathbf{X}^{(k)}$ be the unfolding that groups the first k tensor indices into the row index and the remaining indices into the column index. Its dimensions and the TT ranks are

$$\begin{aligned} \mathbf{X}^{(k)} &\in \mathbb{R}^{(n_1 \cdots n_k) \times (n_{k+1} \cdots n_d)}, \\ r_k &:= \text{rank}(\mathbf{X}^{(k)}), \quad \text{rank}_{\text{TT}}(\mathcal{X}) := (r_1, \dots, r_{d-1}). \end{aligned}$$

These unfolding ranks are the minimal internal dimensions of an exact TT representation. For a rank vector $\mathbf{r} = (r_1, \dots, r_{d-1})$, the inequality $\text{rank}_{\text{TT}}(\mathcal{X}) \leq \mathbf{r}$ is understood componentwise; a scalar bound r means that every TT rank is at most r .

The TT representation stores $\sum_{k=1}^d n_k r_{k-1} r_k$ parameters rather than the $\prod_{k=1}^d n_k$ entries of the full tensor. If $n_k \leq n$ and $r_k \leq r$, this is $\mathcal{O}(dnr^2)$ storage, or $\mathcal{O}(dr^2)$ in the binary setting.

TT-SVD. TT-SVD constructs the cores sequentially. At step $k = 1, \dots, d-1$, it reshapes the current remainder into a matrix M_k , computes $M_k = U_k \Sigma_k V_k^\top$, reshapes U_k into the k th core, and passes $\Sigma_k V_k^\top$ to the next step; the final remainder is reshaped into $\mathcal{G}_d$. Without truncation, this produces an exact TT representation with the unfolding ranks above.

Given a prescribed rank vector $\mathbf{r} = (r_1, \dots, r_{d-1})$, truncated TT-SVD truncates the k th SVD to rank at most r_k . The resulting approximation, denoted by $\text{TT-SVD}_{\mathbf{r}}(\mathcal{X})$, has TT rank at most $\mathbf{r}$ and satisfies the quasi-optimality estimate [18, Corollary 2.4]:

$$\|\mathcal{X} - \text{TT-SVD}_{\mathbf{r}}(\mathcal{X})\|_F \leq \sqrt{d-1} \inf_{\text{rank}_{\text{TT}}(\mathcal{Y}) \leq \mathbf{r}} \|\mathcal{X} - \mathcal{Y}\|_F.$$

This truncated map is used in the proposed algorithms.

2.3 Low-order interaction space

We first introduce a tensor-product basis for $\mathbb{R}^{2 \times \dots \times 2}$. Define the two local basis vectors

$$\mathbf{e}_0 := \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad \mathbf{e}_1 := \begin{bmatrix} 0 \\ 1 \end{bmatrix}. \quad (13)$$

For each $S \subseteq [d]$, define

$$\mathcal{E}_S := \mathbf{e}_1^{(S)} \circ \dots \circ \mathbf{e}_d^{(S)}, \quad \mathbf{e}_j^{(S)} := \begin{cases} \mathbf{e}_1, & j \in S, \\ \mathbf{e}_0, & j \notin S, \end{cases} \quad j = 1, \dots, d. \quad (14)$$

The family $\{\mathcal{E}_S : S \subseteq [d]\}$ is the canonical orthonormal basis of $\mathbb{R}^{2 \times \dots \times 2}$, indexed by subsets $S \subseteq [d]$ of the binary modes.

The connection with low-order interactions becomes explicit after applying the Walsh–Hadamard transform. The local factors satisfy

$$\mathbf{H}_2 \mathbf{e}_0 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad \mathbf{H}_2 \mathbf{e}_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix}. \quad (15)$$

Since $\mathcal{H}$ acts modewise,

$$\mathcal{H}\mathcal{E}_S = \left(\mathbf{H}_2 \mathbf{e}_1^{(S)} \right) \circ \dots \circ \left(\mathbf{H}_2 \mathbf{e}_d^{(S)} \right). \quad (16)$$

The transformed tensor $\mathcal{H}\mathcal{E}_S$ is the basis function associated with the S -interaction in the ANOVA decomposition under the uniform product measure. Indeed, $\mathbf{H}_2 \mathbf{e}_0$ is constant, whereas $\mathbf{H}_2 \mathbf{e}_1$ has zero average over the two binary values. Therefore, $\mathcal{H}\mathcal{E}_S$ is constant outside S and has zero marginal mean in S , as required for a pure S -interaction [31, 19].

Definition 2.1 (Order- s interaction space). For $0 \leq s \leq d$, define

$$\mathcal{V}_s := \left\{ \sum_{\substack{S \subseteq [d] \\ |S| \leq s}} c_S \mathcal{E}_S \mid c_S \in \mathbb{R} \right\}. \quad (17)$$

Because the cluster basis tensors are orthonormal, the expansion in the definition is unique. In the tensor domain, $\mathcal{V}_s$ contains only cluster-basis coefficients indexed by sets of cardinality at most s ; in the spatial domain, $\mathcal{H}(\mathcal{V}_s)$ is precisely the order- s truncated ANOVA space. Such low-order interaction structure arises in many physical and scientific applications, where high-dimensional functions are governed predominantly by interactions among small subsets of variables [25]. This construction is also closely related to the low-order cluster bases used in the atomic cluster expansion [6] and in recent work on tensor-train density estimation [21] and operator learning [14].

For the vectorized formulation, let

$$V_s := \text{vec}(\mathcal{V}_s) \subseteq \mathbb{R}^{2^d}. \quad (18)$$

We denote the orthogonal projectors onto the tensor and vectorized spaces by

$$\mathcal{P}_{\mathcal{V}_s} : \mathbb{R}^{2 \times \dots \times 2} \rightarrow \mathcal{V}_s, \quad P_{V_s} : \mathbb{R}^{2^d} \rightarrow V_s, \quad (19)$$

respectively. They are related through vectorization:

$$\text{vec}(\mathcal{P}_{\mathcal{V}_s} \mathcal{X}) = P_{V_s} \text{vec}(\mathcal{X}). \quad (20)$$

Remark 2.2 (Dimension of the interaction space). Since the basis tensors are indexed by subsets of cardinality at most s ,

$$D_s := \dim(\mathcal{V}_s) = \dim(V_s) = \sum_{\ell=0}^s \binom{d}{\ell}. \quad (21)$$

For $1 \leq s \leq d$,

$$D_s \leq \left(\frac{ed}{s} \right)^s, \quad (22)$$

which is the standard combinatorial bound; see, e.g., [7]. Thus, for fixed s , the dimension grows polynomially in d , in contrast to the full dimension 2^d .

2.4 Structured recovery model

We now formalize the structured recovery problem. The unknown vector and its binary tensorization are

$$\mathbf{x}^* \in \mathbb{R}^N, \quad \mathcal{X}^* := \text{ten}(\mathbf{x}^*) \in \mathbb{R}^{2 \times \cdots \times 2}. \quad (23)$$

Binary tensorization is an isometric reindexing, so recovering $\mathbf{x}^*$ is equivalent to recovering $\mathcal{X}^*$. We therefore formulate the structural model and recovery algorithm in tensor form, where TT rank and sparse low-order interaction structure are explicit, and pass to vector form only for sampled Hadamard-row calculations. After recovering $\mathcal{X}^*$, we obtain the original vector via $\mathbf{x}^* = \text{vec}(\mathcal{X}^*)$.

We impose two assumptions on the tensorization. First, $\mathcal{X}^*$ has bounded TT rank:

$$\text{rank}_{\text{TT}}(\mathcal{X}^*) \leq r. \quad (24)$$

Second, we assume that the cluster-basis expansion can be written as a sum of at most k components, each involving at most s binary modes. Specifically, we define

$$\text{LC}_k(s) := \left\{ \sum_{t=1}^k \mathcal{X}_{S_t} \mid \begin{array}{l} S_t \subseteq [d], \quad |S_t| \leq s, \\ \mathcal{X}_{S_t} \in \text{span}\{\mathcal{E}_I : I \subseteq S_t\}, \quad t = 1, \dots, k \end{array} \right\}. \quad (25)$$

Each $\mathcal{X}_{S_t}$ is supported on coordinates indexed by $I \subseteq S_t$, so it has at most $2^{|S_t|} \leq 2^s$ nonzero entries. Hence every tensor in $\text{LC}_k(s)$ has at most $k2^s$ nonzero cluster-basis coefficients, and $\text{LC}_k(s)$ is a sparse low-order model contained in $\mathcal{V}_s$. The component decomposition need not be unique because these coordinate supports may overlap.

Combining the two assumptions, we define the model class

$$\mathfrak{M}_{r,s,k}(C) := \{ \mathcal{X} \in \mathbb{R}^{2 \times \cdots \times 2} : \text{rank}_{\text{TT}}(\mathcal{X}) \leq r, \mathcal{X} \in \text{LC}_k(s), \|\mathcal{X}\|_F \leq C \}. \quad (26)$$

Here, r bounds the TT ranks; s and k separately control the maximum interaction order and the number of active interaction components; and C bounds the signal scale through the Frobenius norm.

We now formalize the measurement model. Draw $\nu_1, \dots, \nu_m$ independently and uniformly from $[N]$ with replacement, and define $\mathbf{S} \in \mathbb{R}^{m \times N}$ by

$$[\mathbf{S}\mathbf{z}]_\ell := z_{\nu_\ell}, \quad \mathbf{z} \in \mathbb{R}^N, \quad \ell \in [m]. \quad (27)$$

For $\mathcal{Z} \in \mathbb{R}^{2 \times \cdots \times 2}$, set $\mathcal{S}(\mathcal{Z}) := \mathbf{S} \text{vec}(\mathcal{Z})$ and define

$$\mathbf{A} := \sqrt{\frac{N}{m}} \mathbf{S} \mathbf{H}, \quad \mathcal{A}(\mathcal{Z}) := \mathbf{A} \text{vec}(\mathcal{Z}) = \sqrt{\frac{N}{m}} \mathcal{S}(\mathcal{H}(\mathcal{Z})). \quad (28)$$

To simplify the subsequent analysis, rescale the Hadamard rows to have unit-magnitude entries:

$$\mathbf{h}_j := \sqrt{N} \mathbf{H}_{j,:}^\top \in \{\pm 1\}^N, \quad j \in [N]. \quad (29)$$

Thus, for $\mathbf{z} = \text{vec}(\mathcal{Z})$,

$$[\mathcal{A}(\mathcal{Z})]_\ell = [\mathbf{A}\mathbf{z}]_\ell = \frac{1}{\sqrt{m}} \langle \mathbf{h}_{\nu_\ell}, \mathbf{z} \rangle_2, \quad \ell \in [m]. \quad (30)$$

The normalized noiseless model is

$$\mathbf{y} = \mathbf{A}\mathbf{x}^* = \mathcal{A}(\mathcal{X}^*). \quad (31)$$

The scaling makes $\mathbf{A}$ isotropic, since $\mathbb{E}[\mathbf{S}^* \mathbf{S}] = (m/N) \mathbf{I}_N$ and hence $\mathbb{E}[\mathbf{A}^* \mathbf{A}] = \mathbf{I}_N$. Given $\mathbf{y}$ and the sampled indices $\nu_1, \dots, \nu_m$, the objective is to recover $\mathcal{X}^* \in \mathfrak{M}_{r,s,k}(C)$.

3 Algorithm

3.1 Alternating linear scheme

We recover the tensorization $\mathcal{X}^*$ in the TT format using the alternating linear scheme (ALS). For a prescribed TT-rank vector $\mathbf{r} = (r_1, \dots, r_{d-1})$, write the current iterate as

$$\mathcal{X} = [\mathcal{G}_1, \dots, \mathcal{G}_d], \quad \mathcal{G}_j \in \mathbb{R}^{r_{j-1} \times n_j \times r_j}, \quad r_0 = r_d = 1. \quad (32)$$

In the binary setting considered here, $n_j = 2$ for every j . The fixed-rank least-squares problem is

$$\min_{\mathcal{G}_1, \dots, \mathcal{G}_d} \frac{1}{2} \|\mathcal{A}([\mathcal{G}_1, \dots, \mathcal{G}_d]) - \mathbf{y}\|_2^2. \quad (33)$$

ALS updates one core while holding all other cores fixed. When updating the q -th core, define the linear insertion map

$$\mathcal{L}_q(\mathcal{Z}) := [\mathcal{G}_1, \dots, \mathcal{G}_{q-1}, \mathcal{Z}, \mathcal{G}_{q+1}, \dots, \mathcal{G}_d] \quad (34)$$

and the induced core-to-measurement operator

$$\mathcal{A}_q(\mathcal{Z}) := \mathcal{A}(\mathcal{L}_q(\mathcal{Z})). \quad (35)$$

Because the TT contraction is linear in any single core, $\mathcal{A}_q$ is linear. The q -th core update is therefore

$$\min_{\mathcal{Z} \in \mathbb{R}^{r_{q-1} \times n_q \times r_q}} \frac{1}{2} \|\mathcal{A}_q(\mathcal{Z}) - \mathbf{y}\|_2^2. \quad (36)$$

Let $\mathbf{z} := \text{vec}(\mathcal{Z})$. The matrix representation of $\mathcal{A}_q$ is the matrix

$$\mathbf{A}_q \in \mathbb{R}^{m \times (r_{q-1} n_q r_q)} \quad \text{such that} \quad \mathcal{A}_q(\mathcal{Z}) = \mathbf{A}_q \mathbf{z}. \quad (37)$$

Consequently, (36) reduces to

$$\min_{\mathbf{z}} \frac{1}{2} \|\mathbf{A}_q \mathbf{z} - \mathbf{y}\|_2^2. \quad (38)$$

Its normal equations are

$$\mathbf{A}_q^* \mathbf{A}_q \mathbf{z} = \mathbf{A}_q^* \mathbf{y}. \quad (39)$$

We solve each least-squares subproblem through its normal equations and reshape the solution into the updated core. ALS then proceeds to the next core. One complete update of all d cores constitutes an ALS sweep.

3.2 Structure-aware initialization

ALS can be sensitive to its initial TT cores and may converge to a poor stationary point. We therefore construct a data-informed initialization using the two structural assumptions of the model.

The construction is motivated by the restricted isometry property (RIP). For tensors $\mathcal{X}$ in the structured model class,

$$\|\mathcal{A}(\mathcal{X})\|_2^2 = \langle \mathcal{A}^* \mathcal{A}(\mathcal{X}), \mathcal{X} \rangle_F \approx \|\mathcal{X}\|_F^2. \quad (40)$$

Thus, heuristically, the quadratic form of $\mathcal{A}^* \mathcal{A}$ is close to that of the identity on the model class. Given $\mathbf{y} = \mathcal{A}(\mathcal{X}^*)$, this motivates the backprojection

$$\mathcal{A}^* \mathbf{y} = \mathcal{A}^* \mathcal{A}(\mathcal{X}^*) \quad (41)$$

as an initial approximation to $\mathcal{X}^*$.

Since $\mathcal{A} = \sqrt{N/m} \mathcal{S} \mathcal{H}$ and the normalized Walsh–Hadamard transform is self-adjoint, its adjoint is

$$\mathcal{A}^* = \sqrt{\frac{N}{m}} \mathcal{H} \mathcal{S}^*. \quad (42)$$

The backprojection need not belong to the low-order interaction space, so we project it onto $\mathcal{V}_s$:

$$\tilde{\mathcal{X}}_0 := \mathcal{P}_{\mathcal{V}_s}(\mathcal{A}^* \mathbf{y}) \in \mathcal{V}_s. \quad (43)$$

We then apply truncated TT-SVD both to construct the TT representation required by ALS and to impose the prescribed TT-rank bound:

$$\mathcal{X}_{\text{init}} = [\mathcal{G}_1^{(0)}, \dots, \mathcal{G}_d^{(0)}] := \text{TT-SVD}_{\mathbf{r}}(\tilde{\mathcal{X}}_0), \quad \text{rank}_{\text{TT}}(\mathcal{X}_{\text{init}}) \leq \mathbf{r}. \quad (44)$$

The resulting cores initialize ALS. The procedure is summarized in Algorithm 1.

Algorithm 1 Structure-aware initialization

Input: Measurements $\mathbf{y}$, adjoint measurement operator $\mathcal{A}^*$, interaction order s , and target TT-rank vector $\mathbf{r}$

- 1: $\tilde{\mathcal{X}}_0 \leftarrow \mathcal{P}_{\mathcal{V}_s}(\mathcal{A}^* \mathbf{y})$ ▷ Low-order projection
- 2: $[\mathcal{G}_1^{(0)}, \dots, \mathcal{G}_d^{(0)}] \leftarrow \text{TT-SVD}_{\mathbf{r}}(\tilde{\mathcal{X}}_0)$ ▷ TT representation with prescribed ranks

Output: Initial TT cores $\mathcal{G}_1^{(0)}, \dots, \mathcal{G}_d^{(0)}$

Since ALS keeps the prescribed ranks fixed, the end-to-end recovery experiments use the oracle choice $\mathbf{r} = \text{rank}_{\text{TT}}(\mathcal{X}^*)$. The initialization-scaling experiments instead use the uniform rank bound $r_j = k$, $j = 1, \dots, d - 1$.

4 Restricted isometry and uniqueness

4.1 RIP for the structured model class

We first prove concentration for a fixed tensor, using the Hadamard-row representation in (30), and then extend the result uniformly by a covering argument.

Lemma 4.1 (Fixed-tensor concentration). *Let $\mathcal{X} \in \text{LC}_k(s) \setminus \{0\}$ be fixed. For every $\varepsilon \in (0, 1)$,*

$$\mathbb{P}(|\|\mathcal{A}(\mathcal{X})\|_2^2 - \|\mathcal{X}\|_F^2| \geq \varepsilon \|\mathcal{X}\|_F^2) \leq 2 \exp\left(-\frac{m\varepsilon^2}{4k2^s}\right). \quad (45)$$

Proof. Let $\mathbf{x} = \text{vec}(\mathcal{X})$ and define

$$Z_\ell := |\langle \mathbf{h}_{\nu_\ell}, \mathbf{x} \rangle|^2, \quad \ell = 1, \dots, m. \quad (46)$$

By (30),

$$\|\mathcal{A}(\mathcal{X})\|_2^2 = \frac{1}{m} \sum_{\ell=1}^m Z_\ell. \quad (47)$$

The variables $Z_1, \dots, Z_m$ are independent and identically distributed. Since $\mathbf{H}$ is orthogonal,

$$\mathbb{E} Z_\ell = \frac{1}{N} \sum_{j=1}^N |\langle \mathbf{h}_j, \mathbf{x} \rangle|^2 = \frac{1}{N} \|\sqrt{N} \mathbf{H} \mathbf{x}\|_2^2 = \|\mathbf{H} \mathbf{x}\|_2^2 = \|\mathbf{x}\|_2^2 = \|\mathcal{X}\|_F^2. \quad (48)$$

Because $\mathcal{X} \in \text{LC}_k(s)$, the support of $\mathbf{x}$ is contained in a set $T \subseteq [N]$ with $|T| \leq k2^s$. Since the entries of every $\mathbf{h}_j$ have magnitude one, the Cauchy-Schwarz inequality gives

$$Z_\ell = \left| \sum_{j \in T} (\mathbf{h}_{\nu_\ell})_j x_j \right|^2 \leq |T| \|\mathbf{x}\|_2^2 \leq k2^s \|\mathcal{X}\|_F^2. \quad (49)$$

Set $B := k2^s \|\mathcal{X}\|_F^2$. Then $0 \leq Z_\ell \leq B$ and $|Z_\ell - \mathbb{E}Z_\ell| \leq B$. Moreover,

$$\text{Var}(Z_\ell) \leq \mathbb{E}Z_\ell^2 \leq B \mathbb{E}Z_\ell = k2^s \|\mathcal{X}\|_F^4. \quad (50)$$

Bernstein's inequality applied to $Z_\ell - \mathbb{E}Z_\ell$ therefore yields

$$\begin{aligned} & \mathbb{P} \left(\left| \frac{1}{m} \sum_{\ell=1}^m Z_\ell - \|\mathcal{X}\|_F^2 \right| \geq \varepsilon \|\mathcal{X}\|_F^2 \right) \\ & \leq 2 \exp \left(-\frac{m\varepsilon^2}{2(1 + \varepsilon/3)k2^s} \right) \leq 2 \exp \left(-\frac{m\varepsilon^2}{4k2^s} \right), \end{aligned} \quad (51)$$

where the last inequality uses $\varepsilon < 1$. $\square$

We next recall the TT covering estimate used in the uniform argument. It follows by covering a suitably orthogonalized collection of TT cores; see, for example, [26].

Lemma 4.2 (Covering number for TT tensors). *Let $d \geq 2$, $r \geq 1$, and $C > 0$, and define*

$$\mathcal{G}_r(C) := \{ \mathcal{X} \in \mathbb{R}^{2 \times \dots \times 2} : \text{rank}_{\text{TT}}(\mathcal{X}) \leq r, \|\mathcal{X}\|_F \leq C \}. \quad (52)$$

There exists an absolute constant $K > 0$ such that, for every $0 < \eta \leq C$,

$$\mathcal{N}(\mathcal{G}_r(C), \|\cdot\|_F, \eta) \leq \left(\frac{Kd\sqrt{r}C}{\eta} \right)^{2dr^2}. \quad (53)$$

We now combine the fixed-tensor estimate with this covering bound.

Theorem 4.3 (Uniform RIP over the structured tensor class). *Let $\varepsilon, \delta \in (0, 1)$. If*

$$m \geq \frac{16k2^s}{\varepsilon^2} \left[\log \frac{2}{\delta} + 2dr^2 \log \left(\frac{16Kd\sqrt{r}k2^s}{\varepsilon} \right) \right], \quad (54)$$

then, with probability at least $1 - \delta$,

$$\left| \|\mathcal{A}(\mathcal{X})\|_2^2 - \|\mathcal{X}\|_F^2 \right| \leq \varepsilon \|\mathcal{X}\|_F^2 \quad \text{for every } \mathcal{X} \in \mathfrak{M}_{r,s,k}(C). \quad (55)$$

Proof. Because scalar multiplication preserves both TT rank and membership in $\text{LC}_k(s)$, it is enough to work on the normalized structured set

$$\mathbb{S}_{r,s,k} := \{ \mathcal{X} \in \mathbb{R}^{2 \times \dots \times 2} : \text{rank}_{\text{TT}}(\mathcal{X}) \leq r, \mathcal{X} \in \text{LC}_k(s), \|\mathcal{X}\|_F = 1 \}. \quad (56)$$

Since $\mathbb{S}_{r,s,k} \subseteq \mathcal{G}_r(1)$, there exists an η -net $\mathcal{N}_\eta \subseteq \mathbb{S}_{r,s,k}$ satisfying

$$|\mathcal{N}_\eta| \leq \mathcal{N} \left(\mathcal{G}_r(1), \|\cdot\|_F, \frac{\eta}{2} \right) \leq \left(\frac{2Kd\sqrt{r}}{\eta} \right)^{2dr^2}. \quad (57)$$

Fix $\mathcal{X} \in \mathbb{S}_{r,s,k}$ and choose $\mathcal{X}^{(p)} \in \mathcal{N}_\eta$ such that $\|\mathcal{X} - \mathcal{X}^{(p)}\|_F \leq \eta$. Let $\mathbf{x}$ and $\mathbf{x}^{(p)}$ denote their vectorizations. The triangle inequality gives

$$\left| \|\mathcal{A}(\mathcal{X})\|_2^2 - 1 \right| \leq \left| \|\mathcal{A}(\mathcal{X})\|_2^2 - \|\mathcal{A}(\mathcal{X}^{(p)})\|_2^2 \right| + \left| \|\mathcal{A}(\mathcal{X}^{(p)})\|_2^2 - 1 \right|. \quad (58)$$

The second term is the fixed-tensor RIP error at the net point. We bound the first term deterministically.

Net approximation error. By (30),

$$\begin{aligned} & \left| \|\mathcal{A}(\mathcal{X})\|_2^2 - \|\mathcal{A}(\mathcal{X}^{(p)})\|_2^2 \right| \\ & \leq \frac{1}{m} \sum_{\ell=1}^m \left| \left\langle \mathbf{h}_{\nu_\ell}, \mathbf{x} - \mathbf{x}^{(p)} \right\rangle \right| \left| \left\langle \mathbf{h}_{\nu_\ell}, \mathbf{x} + \mathbf{x}^{(p)} \right\rangle \right|. \end{aligned} \quad (59)$$

Each of $\mathbf{x}$ and $\mathbf{x}^{(p)}$ has at most $k2^s$ nonzero entries. Their sum and difference are therefore supported on at most $2k2^s$ coordinates. Moreover, $\|\mathbf{x} + \mathbf{x}^{(p)}\|_2 \leq 2$ because both tensors belong to $\mathbb{S}_{r,s,k}$. Thus, for every sampled row,

$$\begin{aligned} \left| \left\langle \mathbf{h}_{\nu_\ell}, \mathbf{x} - \mathbf{x}^{(p)} \right\rangle \right| & \leq \sqrt{2k2^s} \eta, \\ \left| \left\langle \mathbf{h}_{\nu_\ell}, \mathbf{x} + \mathbf{x}^{(p)} \right\rangle \right| & \leq 2\sqrt{2k2^s}. \end{aligned} \quad (60)$$

Consequently,

$$\left| \|\mathcal{A}(\mathcal{X})\|_2^2 - \|\mathcal{A}(\mathcal{X}^{(p)})\|_2^2 \right| \leq 4k2^s \eta. \quad (61)$$

Choice of η and union bound. Choose

$$\eta := \frac{\varepsilon}{8k2^s}. \quad (62)$$

The net approximation error is then at most $\varepsilon/2$, and (57) gives

$$|\mathcal{N}_\eta| \leq \left(\frac{16Kd\sqrt{r}k2^s}{\varepsilon} \right)^{2dr^2}. \quad (63)$$

Lemma 4.1, applied with accuracy $\varepsilon/2$ to each $\mathcal{X}^{(p)} \in \mathcal{N}_\eta$, gives

$$\mathbb{P} \left(\left| \|\mathcal{A}(\mathcal{X}^{(p)})\|_2^2 - 1 \right| \geq \frac{\varepsilon}{2} \right) \leq 2 \exp \left(-\frac{m\varepsilon^2}{16k2^s} \right). \quad (64)$$

Taking a union bound over $\mathcal{N}_\eta$, the failure probability is at most

$$2 \left(\frac{16Kd\sqrt{r}k2^s}{\varepsilon} \right)^{2dr^2} \exp \left(-\frac{m\varepsilon^2}{16k2^s} \right). \quad (65)$$

By (54), this probability is at most δ .

Therefore, with probability at least $1 - \delta$, every net point satisfies

$$\left| \|\mathcal{A}(\mathcal{X}^{(p)})\|_2^2 - 1 \right| \leq \frac{\varepsilon}{2}. \quad (66)$$

Combining this with the deterministic approximation estimate gives, for every $\mathcal{X} \in \mathbb{S}_{r,s,k}$,

$$\left| \|\mathcal{A}(\mathcal{X})\|_2^2 - 1 \right| \leq \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon. \quad (67)$$

Finally, for any nonzero $\mathcal{X} \in \mathfrak{M}_{r,s,k}(C)$, the normalized tensor $\mathcal{X}/\|\mathcal{X}\|_F$ belongs to $\mathbb{S}_{r,s,k}$. Homogeneity gives the claimed estimate. The zero tensor satisfies it trivially. $\square$

Remark 4.4 (Comparison with general sparse RIP bounds). Let $q := k2^s$. Then the vectorization of every tensor in $\mathfrak{M}_{r,s,k}(C)$ is q -sparse in $\mathbb{R}^N$. For fixed ε and δ , using $d = \log_2 N$ in Theorem 4.3 gives $m = \mathcal{O}(qr^2 \log N \log(\sqrt{r} q \log N))$. For bounded TT rank, this becomes

$$m = \mathcal{O}(q \log N \log(q \log N)). \quad (68)$$

For comparison, if the tensor structure is ignored, classical RIP results for arbitrary q -sparse vectors apply. Rudelson and Vershynin showed that, for an orthogonal matrix with entries bounded by $O(N^{-1/2})$,

$$m = \mathcal{O}(q \log N \log^2 q \log(q \log N)) = \mathcal{O}(q \log^4 N) \quad (69)$$

measurements suffice [27]. The normalized Walsh–Hadamard matrix belongs to this class. For bounded-entry unitary matrices with independent uniform row sampling, as used here, Haviv and Regev sharpened the requirement to

$$m = \mathcal{O}(q \log^2 q \log N) \quad (70)$$

measurements [10]. For bounded TT rank, our model-aware estimate removes the factor $\log^2 q$ from the Rudelson–Vershynin bound. It does not uniformly improve on the Haviv–Regev bound: the remaining factors are $\log(q \log N)$ in our estimate and $\log^2 q$ in theirs, so their ordering depends on how q scales with N . Nevertheless, for the smaller family determined by the TT-rank and interaction constraints, a single-scale covering argument—without a multiscale chaining construction—already yields a bound of comparable polylogarithmic order.

Remark 4.5 (Extension to the full order- s interaction space $\mathcal{V}_s$). In practice, the number k of active low-order interaction components may be unknown; one may only know that $\mathcal{X} \in \mathcal{V}_s$ has TT rank at most r . For every $\mathcal{X} \in \mathcal{V}_s$, the support of $\text{vec}(\mathcal{X})$ is contained in the coordinates indexed by $S \subseteq [d]$ with $|S| \leq s$, and hence has cardinality at most

$$D_s = \sum_{\ell=0}^s \binom{d}{\ell}.$$

Repeating the proof of Theorem 4.3 with this support bound in place of $k2^s$, it is sufficient to require

$$m \geq \frac{16D_s}{\varepsilon^2} \left[\log \frac{2}{\delta} + 2dr^2 \log \left(\frac{8Kd\sqrt{r} D_s}{\varepsilon} \right) \right]. \quad (71)$$

Using $D_s \leq (ed/s)^s$ from Remark 2.2, it is enough to require

$$m \geq \frac{16}{\varepsilon^2} \left(\frac{ed}{s} \right)^s \left[\log \frac{2}{\delta} + 2dr^2 \log \left(\frac{8Kd\sqrt{r}}{\varepsilon} \left(\frac{ed}{s} \right)^s \right) \right]. \quad (72)$$

For fixed s , this condition remains polynomial in d . Thus, extending the active-component model to the full interaction space $\mathcal{V}_s$ preserves the qualitative sample-complexity scaling.

4.2 Uniqueness of noiseless recovery

We next explain how a structured RIP implies uniqueness of the noiseless constrained recovery problem. Let $\mathcal{X}^* \in \mathfrak{M}_{r,s,k}(C)$ and consider

$$\hat{\mathcal{X}} \in \arg \min_{\mathcal{X} \in \mathfrak{M}_{r,s,k}(C)} \|\mathcal{A}(\mathcal{X}) - \mathbf{y}\|_2, \quad \mathbf{y} = \mathcal{A}(\mathcal{X}^*). \quad (73)$$

Corollary 4.6 (Uniqueness under structured RIP). *Suppose that $\mathcal{A}$ satisfies the RIP on $\mathfrak{M}_{2r,s,2k}(2C)$ with a constant $\varepsilon < 1$; that is,*

$$(1 - \varepsilon)\|\mathcal{D}\|_F^2 \leq \|\mathcal{A}(\mathcal{D})\|_2^2 \leq (1 + \varepsilon)\|\mathcal{D}\|_F^2 \quad \text{for every } \mathcal{D} \in \mathfrak{M}_{2r,s,2k}(2C). \quad (74)$$

Then $\mathcal{X}^*$ is the unique minimizer of (73).

Proof. The ground-truth tensorization is feasible and gives zero residual, so every minimizer $\hat{\mathcal{X}}$ satisfies $\mathcal{A}(\hat{\mathcal{X}}) = \mathcal{A}(\mathcal{X}^*)$. Set $\mathcal{D} := \hat{\mathcal{X}} - \mathcal{X}^*$. Since both tensors belong to $\mathfrak{M}_{r,s,k}(C)$, subadditivity of the ranks of their TT unfoldings gives $\text{rank}_{\text{TT}}(\mathcal{D}) \leq 2r$. Concatenating their two cluster-component lists and negating the coefficients of $\mathcal{X}^*$ shows that $\mathcal{D} \in \text{LC}_{2k}(s)$. Moreover, $\|\mathcal{D}\|_F \leq \|\hat{\mathcal{X}}\|_F + \|\mathcal{X}^*\|_F \leq 2C$. Thus, $\mathcal{D} \in \mathfrak{M}_{2r,s,2k}(2C)$, and the lower RIP inequality gives

$$(1 - \varepsilon)\|\mathcal{D}\|_F^2 \leq \|\mathcal{A}(\mathcal{D})\|_2^2 = 0. \quad (75)$$

Since $\varepsilon < 1$, we obtain $\mathcal{D} = 0$, and hence $\hat{\mathcal{X}} = \mathcal{X}^*$. $\square$

Consequently, for any $\delta \in (0, 1)$, applying Theorem 4.3 to $\mathfrak{M}_{2r,s,2k}(2C)$ shows that uniqueness holds with probability at least $1 - \delta$, provided (54) holds with r and k replaced by $2r$ and $2k$.

5 Analysis of the structure-aware initialization

Algorithm 1 proceeds in two stages:

$$\begin{aligned} \tilde{\mathcal{X}}_0 &= \mathcal{P}_{\mathcal{V}_s} \mathcal{A}^* \mathcal{A}(\mathcal{X}^*), \\ \mathcal{X}_{\text{init}} &= \text{TT-SVD}_{\mathbf{r}}(\tilde{\mathcal{X}}_0). \end{aligned} \quad (76)$$

Accordingly, we first bound the projected backprojection error and then control the deterministic error introduced by TT-SVD truncation.

5.1 Projected backprojection error

Let $\mathcal{I}$ denote the identity operator on $\mathbb{R}^{2 \times \dots \times 2}$. Since $\mathfrak{M}_{r,s,k}(C) \subseteq \mathcal{V}_s$, every $\mathcal{X} \in \mathfrak{M}_{r,s,k}(C)$ satisfies $\mathcal{P}_{\mathcal{V}_s}(\mathcal{X}) = \mathcal{X}$, and hence

$$\mathcal{P}_{\mathcal{V}_s} \mathcal{A}^* \mathcal{A}(\mathcal{X}) - \mathcal{X} = \mathcal{P}_{\mathcal{V}_s} (\mathcal{A}^* \mathcal{A} - \mathcal{I}) \mathcal{P}_{\mathcal{V}_s}(\mathcal{X}). \quad (77)$$

Define the compressed error operator $\mathcal{B}_s := \mathcal{P}_{\mathcal{V}_s} (\mathcal{A}^* \mathcal{A} - \mathcal{I}) \mathcal{P}_{\mathcal{V}_s}$. The TT-RIP argument used by Qin, Wakin, and Zhu [23] for the unprojected spectral initializer cannot be applied directly here. Controlling $\mathcal{B}_s$ requires operator-norm control of $\mathcal{P}_{\mathcal{V}_s} \mathcal{A}^* \mathcal{A} \mathcal{P}_{\mathcal{V}_s}$ on the full space $\mathcal{V}_s$, whereas Theorem 4.3 gives RIP only on $\mathfrak{M}_{r,s,k}(C) \subseteq \mathcal{V}_s$. Applying the same argument would therefore require an RIP on the larger space $\mathcal{V}_s$, which is not provided by Theorem 4.3. We instead analyze $\mathcal{B}_s$ directly using matrix concentration. It follows that

$$\sup_{\substack{\mathcal{X} \in \mathfrak{M}_{r,s,k}(C) \\ \mathcal{X} \neq 0}} \frac{\|\mathcal{P}_{\mathcal{V}_s} \mathcal{A}^* \mathcal{A}(\mathcal{X}) - \mathcal{X}\|_F}{\|\mathcal{X}\|_F} \leq \|\mathcal{B}_s\|_{F \rightarrow F}, \quad (78)$$

where $\|\cdot\|_{F \rightarrow F}$ is the operator norm on $\mathcal{V}_s$ induced by the Frobenius norm. We first establish a high-probability bound for this operator norm using matrix Bernstein's inequality.

Lemma 5.1 (Compressed error operator bound). *For every $\varepsilon \in (0, 1)$,*

$$\mathbb{P}(\|\mathcal{B}_s\|_{F \rightarrow F} \geq \varepsilon) \leq 2D_s \exp\left(-\frac{3m\varepsilon^2}{8D_s}\right). \quad (79)$$

Proof. Under vectorization, $\mathcal{B}_s$ is represented on V_s by the self-adjoint matrix

$$\mathbf{B}_s := P_{V_s}(\mathbf{A}^* \mathbf{A} - \mathbf{I}_N)P_{V_s}. \quad (80)$$

Since vectorization is an isometry, $\|\mathcal{B}_s\|_{F \rightarrow F} = \|\mathbf{B}_s\|_2$. For each ℓ , define

$$\mathbf{u}_\ell := P_{V_s} \mathbf{h}_{\nu_\ell} \in V_s, \quad \mathbf{Z}_\ell := \mathbf{u}_\ell \mathbf{u}_\ell^* - P_{V_s}. \quad (81)$$

By (30),

$$\mathbf{B}_s = \frac{1}{m} \sum_{\ell=1}^m \mathbf{Z}_\ell. \quad (82)$$

The matrices $\mathbf{Z}_1, \dots, \mathbf{Z}_m$ are independent and self-adjoint.

Uniform sampling and Hadamard orthogonality give

$$\mathbb{E}[\mathbf{u}_\ell \mathbf{u}_\ell^*] = P_{V_s} \left(\frac{1}{N} \sum_{j=1}^N \mathbf{h}_j \mathbf{h}_j^* \right) P_{V_s} = P_{V_s}. \quad (83)$$

Hence $\mathbb{E}\mathbf{Z}_\ell = 0$. Since P_{V_s} retains D_s coordinates and every entry of $\mathbf{h}_{\nu_\ell}$ has magnitude one,

$$\|\mathbf{u}_\ell\|_2^2 = D_s. \quad (84)$$

On V_s , the eigenvalues of $\mathbf{Z}_\ell$ are $D_s - 1$ in the direction of $\mathbf{u}_\ell$ and -1 on its orthogonal complement. Thus, $\|\mathbf{Z}_\ell\|_2 \leq D_s$. Furthermore, since $P_{V_s} \mathbf{u}_\ell = \mathbf{u}_\ell$ and $P_{V_s}^2 = P_{V_s}$,

$$\begin{aligned} \mathbf{Z}_\ell^2 &= (\mathbf{u}_\ell \mathbf{u}_\ell^* - P_{V_s})^2 \\ &= (D_s - 2) \mathbf{u}_\ell \mathbf{u}_\ell^* + P_{V_s}, \\ \mathbb{E}\mathbf{Z}_\ell^2 &= (D_s - 2) P_{V_s} + P_{V_s} \\ &= (D_s - 1) P_{V_s}. \end{aligned} \quad (85)$$

Thus, the variance parameter is $\|\sum_{\ell=1}^m \mathbb{E}\mathbf{Z}_\ell^2\|_2 = m(D_s - 1)$. The self-adjoint matrix Bernstein inequality [32], with threshold $m\varepsilon$, yields

$$\begin{aligned} \mathbb{P}(\|\mathcal{B}_s\|_{F \rightarrow F} \geq \varepsilon) &\leq 2D_s \exp\left(-\frac{m\varepsilon^2}{2(D_s - 1) + (2/3)D_s\varepsilon}\right) \\ &\leq 2D_s \exp\left(-\frac{3m\varepsilon^2}{8D_s}\right), \end{aligned} \quad (86)$$

where the last inequality uses $\varepsilon < 1$. □

Combining Lemma 5.1 with (78) gives the central backprojection guarantee.

Theorem 5.2 (Uniform projected backprojection bound). *Let $\varepsilon, \delta \in (0, 1)$. If*

$$m \geq \frac{8D_s}{3\varepsilon^2} \log\left(\frac{2D_s}{\delta}\right), \quad (87)$$

then, with probability at least $1 - \delta$,

$$\|\mathcal{P}_{V_s} \mathcal{A}^* \mathcal{A}(\mathcal{X}) - \mathcal{X}\|_F \leq \varepsilon \|\mathcal{X}\|_F \quad \text{for every } \mathcal{X} \in \mathfrak{M}_{r,s,k}(C). \quad (88)$$

5.2 TT-SVD truncation and initialization guarantee

The second initialization stage imposes the target TT-rank by truncated TT-SVD. The following deterministic lemma transfers any pre-truncation error bound to the truncated tensor.

Lemma 5.3 (TT-SVD postprocessing error). *Let $\mathcal{X}$ be a d -way tensor satisfying $\text{rank}_{\text{TT}}(\mathcal{X}) \leq \mathbf{r}$, and let $\tilde{\mathcal{X}}$ be any perturbation of $\mathcal{X}$. Define*

$$\hat{\mathcal{X}} := \text{TT-SVD}_{\mathbf{r}}(\tilde{\mathcal{X}}). \quad (89)$$

Then

$$\|\hat{\mathcal{X}} - \mathcal{X}\|_F \leq \left(1 + \sqrt{d-1}\right) \|\tilde{\mathcal{X}} - \mathcal{X}\|_F. \quad (90)$$

Proof. The triangle inequality and TT-SVD quasi-optimality [18, Corollary 2.4] give

$$\begin{aligned} \|\hat{\mathcal{X}} - \mathcal{X}\|_F &\leq \|\hat{\mathcal{X}} - \tilde{\mathcal{X}}\|_F + \|\tilde{\mathcal{X}} - \mathcal{X}\|_F \\ &\leq \sqrt{d-1} \inf_{\text{rank}_{\text{TT}}(\mathcal{Z}) \leq \mathbf{r}} \|\tilde{\mathcal{X}} - \mathcal{Z}\|_F + \|\tilde{\mathcal{X}} - \mathcal{X}\|_F \\ &\leq \left(1 + \sqrt{d-1}\right) \|\tilde{\mathcal{X}} - \mathcal{X}\|_F. \end{aligned} \quad (91)$$

The last inequality uses the admissible choice $\mathcal{Z} = \mathcal{X}$. $\square$

Combining Theorem 5.2 with Lemma 5.3 gives the final initialization guarantee.

Theorem 5.4 (Uniform initialization error bound). *Let $\varepsilon, \delta \in (0, 1)$ and set $\alpha_d := 1 + \sqrt{d-1}$. If*

$$m \geq \frac{8D_s\alpha_d^2}{3\varepsilon^2} \log\left(\frac{2D_s}{\delta}\right), \quad (92)$$

then, with probability at least $1 - \delta$, simultaneously for every $\mathcal{X} \in \mathfrak{M}_{r,s,k}(C)$ and every target rank vector $\mathbf{r}$ satisfying $\text{rank}_{\text{TT}}(\mathcal{X}) \leq \mathbf{r}$, the initializer

$$\mathcal{X}_{\text{init}} = \text{TT-SVD}_{\mathbf{r}}(\mathcal{P}_{\mathcal{V}_s} \mathcal{A}^* \mathcal{A}(\mathcal{X})) \quad (93)$$

satisfies

$$\|\mathcal{X}_{\text{init}} - \mathcal{X}\|_F \leq \varepsilon \|\mathcal{X}\|_F. \quad (94)$$

Proof. Apply Theorem 5.2 with accuracy ε/α_d and then Lemma 5.3. Under (92), with probability at least $1 - \delta$,

$$\|\mathcal{X}_{\text{init}} - \mathcal{X}\|_F \leq \alpha_d \|\mathcal{P}_{\mathcal{V}_s} \mathcal{A}^* \mathcal{A}(\mathcal{X}) - \mathcal{X}\|_F \leq \varepsilon \|\mathcal{X}\|_F. \quad (95)$$

The estimate holds simultaneously for every admissible $\mathcal{X}$ and $\mathbf{r}$. $\square$

Remark 5.5 (Scaling of the initialization guarantee). For $1 \leq s < d$, the bound $D_s \leq (ed/s)^s$ shows that it is sufficient to take

$$m \geq \frac{8\alpha_d^2}{3\varepsilon^2} \left(\frac{ed}{s}\right)^s \log\left[\frac{2}{\delta} \left(\frac{ed}{s}\right)^s\right]. \quad (96)$$

For fixed s, ε , and δ , we have $\alpha_d^2 = \mathcal{O}(d)$; hence, the sufficient sample size satisfies

$$m = \mathcal{O}(d^{s+1} \log d). \quad (97)$$

Thus, for fixed interaction order s , the required number of measurements grows polynomially with the tensor order d , rather than exponentially through the ambient dimension $N = 2^d$.

6 Experiments

We test three aspects of the proposed framework: restricted isometry on the structured model class, the accuracy of the structure-aware initialization, and its effect on the full ALS recovery workflow.

Common experimental setup. All experiments use synthetic data from the same sliding-window interaction model so that the sparse low-order interaction structure and TT ranks are controlled.

Fix an interaction order s , a number of active components k , and a step size p . For $t = 1, \dots, k$, define the sliding-window mode sets

$$a_t := 1 + (t - 1)p, \quad I_t := \{a_t, \dots, a_t + s - 1\}, \quad (98)$$

where $a_t + s - 1 \leq d$. We then construct the target tensor as a sum of rank-one cluster components associated with these windows:

$$\begin{aligned} \mathcal{X}^* &:= \sum_{t=1}^k \mathcal{X}_{I_t}, & \mathcal{X}_{I_t} &:= \mathbf{u}_1^{(t)} \circ \dots \circ \mathbf{u}_d^{(t)}, \\ \mathbf{u}_j^{(t)} &:= \begin{cases} \mathbf{v}_j^{(t)}, & j \in I_t, \\ (1, 0)^\top, & j \notin I_t. \end{cases} \end{aligned} \quad (99)$$

Each active vector is sampled independently as $\mathbf{v}_j^{(t)} = \mathbf{z}_j^{(t)} / \|\mathbf{z}_j^{(t)}\|_2$, where $\mathbf{z}_j^{(t)} \sim \mathcal{N}(0, \mathbf{I}_2)$. By the definition of $\text{LC}_k(s)$, $\mathcal{X}_{I_t}$ is a cluster component involving only the modes in I_t . Thus, $\mathcal{X}^* \in \text{LC}_k(s) \subseteq \mathcal{V}_s$ and $\text{rank}_{\text{TT}}(\mathcal{X}^*) \leq k$. In practice, the TT ranks can be smaller because only active windows crossing a given tensor-train cut contribute to the corresponding unfolding rank. For each tensor $\mathcal{X}$, let $r_{\max}(\mathcal{X}) := \max_j r_j(\mathcal{X})$. For each fixed sliding-window pattern, $r_{\max}(\mathcal{X})$ appeared empirically constant across the generated tensors: its sample mean and sample maximum coincided within each test set.

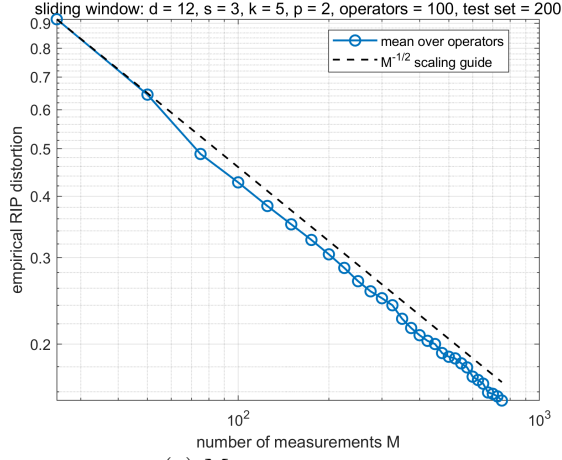
6.1 Tests for the RIP

Theorem 4.3 takes a supremum over the full model class and gives a probabilistic guarantee for the random operator. We investigate it empirically using a finite-test-set approximation to the supremum and repeated independent draws of the operator.

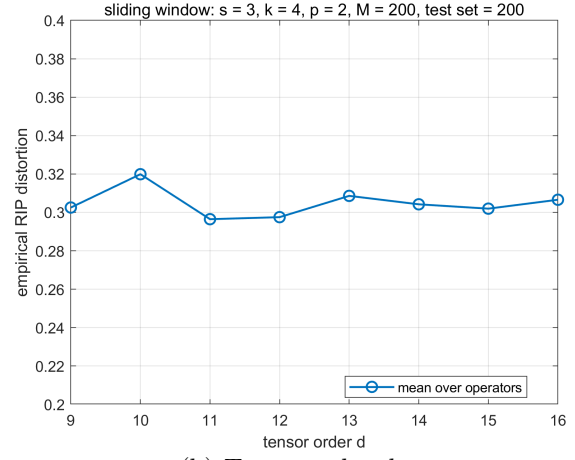
We draw 100 independent operators whose rows are sampled uniformly with replacement. For each operator, we generate an independent test set $\mathcal{T}$ of 200 tensors and estimate its largest relative distortion by

$$\widehat{\delta}_{\mathcal{T}}(\mathcal{A}) := \max_{\mathcal{X} \in \mathcal{T}} \left| \frac{\|\mathcal{A}(\mathcal{X})\|_2^2}{\|\mathcal{X}\|_F^2} - 1 \right|. \quad (100)$$

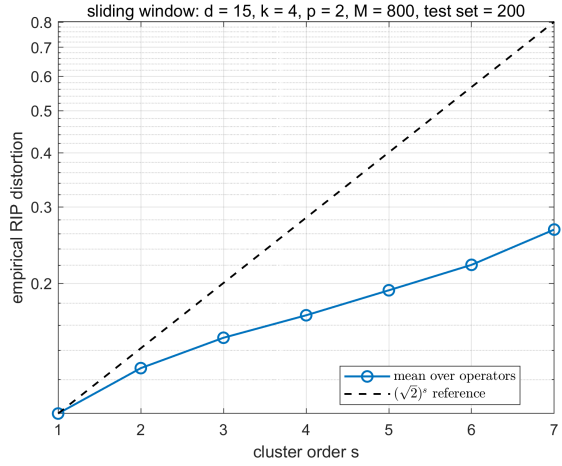
We report the mean of $\widehat{\delta}_{\mathcal{T}}(\mathcal{A})$ over the 100 operators. All operator applications use the fast Walsh–Hadamard transform defined in Section 2.1.



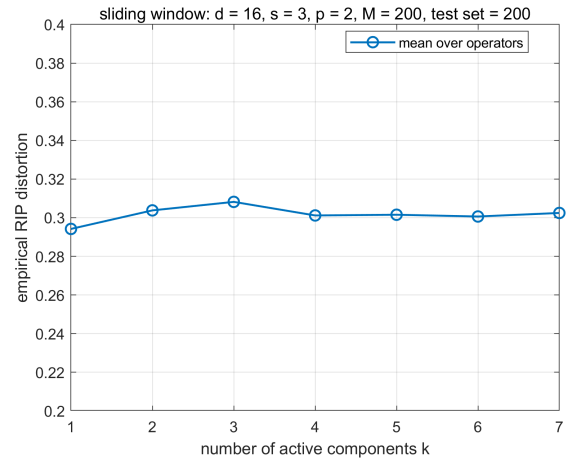
(a) Measurements m .



(b) Tensor order d .



(c) Interaction order s .



(d) Active components k .

Figure 1: Empirical RIP estimates under sweeps of (a) $m = 25, 50, \dots, 750$, (b) $d = 9, \dots, 16$, (c) $s = 1, \dots, 7$, and (d) $k = 1, \dots, 7$.

To extract qualitative predictions from Theorem 4.3, we vary one parameter at a time. Fixing the failure probability and ignoring logarithmic factors and the dependence on the TT-rank bound r , the sample-complexity bound in (54) suggests the following heuristic scalings:

$$\varepsilon_{\text{RIP}} \propto \begin{cases} m^{-1/2}, & \text{as the measurement budget } m \text{ varies,} \\ d^{1/2}, & \text{as the tensor order } d \text{ varies,} \\ (\sqrt{2})^s, & \text{as the interaction order } s \text{ varies,} \\ k^{1/2}, & \text{as the number of active components } k \text{ varies.} \end{cases} \quad (101)$$

The reference curves in panels (a) and (c) are scaling guides, not numerical upper bounds: each is normalized to the first empirical point of the corresponding sweep and indicates only the predicted dependence on the varied parameter.

Figure 1(a) varies the measurement budget $m = 25, 50, \dots, 750$ and fixes $(d, s, k, p) = (12, 3, 5, 2)$. The empirical distortion closely follows the Monte Carlo rate $m^{-1/2}$.

Panel (b) varies the tensor order $d = 9, \dots, 16$ and fixes $(s, k, p, m) = (3, 4, 2, 200)$. The distortion remains between approximately 0.30 and 0.32, with no visible $d^{1/2}$ growth; the observed dependence on d is therefore milder than the leading worst-case prediction, suggesting that the current theory may be conservative.

Panel (c) varies the interaction order $s = 1, \dots, 7$ and fixes $(d, k, p, m) = (15, 4, 2, 800)$. The distortion increases with s , as predicted, but much more slowly than the $(\sqrt{2})^s$ guide. The theorem uses the generic sparsity estimate $|\text{supp}(\text{vec}(\mathcal{X}))| \leq k2^s$, obtained by adding the 2^s possible cluster-basis coordinates of each component. Neighboring windows share some of these coordinates, so the support of their sum can be substantially smaller than $k2^s$. This smaller effective sparsity may partly explain the milder observed growth.

Panel (d) varies the number of active components $k = 1, \dots, 7$ and fixes $(d, s, p, m) = (16, 3, 2, 200)$. The distortion remains close to 0.30, with no visible $k^{1/2}$ growth; the observed dependence on k is again more favorable than the leading worst-case prediction.

Together, the four sweeps support Theorem 4.3 empirically, while indicating that its dependence on d , s , and k is conservative for the sampled sliding-window family. The experiment replaces the supremum over the full model class by a maximum over a finite sliding-window test set, so these results support the predicted scaling but do not verify the uniform guarantee.

6.2 Initialization error and scaling

To isolate the effect of each step in Algorithm 1, we compare three estimates against the ground-truth tensor $\mathcal{X}^*$: the adjoint backprojection $\mathcal{A}^*\mathbf{y}$, its projection onto $\mathcal{V}_s$, and the TT-SVD truncation of that projection to rank $\mathbf{r}$. We also include the standard spectral initializer, obtained by applying the same TT-SVD truncation directly to $\mathcal{A}^*\mathbf{y}$, as a benchmark that omits the model projection. For every estimate $\mathcal{Z}$, we report the relative error $e(\mathcal{Z}) = \|\mathcal{Z} - \mathcal{X}^*\|_F / \|\mathcal{X}^*\|_F$. Because $\mathcal{X}^*$ is a sum of k rank-one cluster components, its TT ranks are at most k . We therefore use the uniform TT-SVD rank cap $r_j = k$, $j = 1, \dots, d-1$. Here, k denotes the number of rank-one cluster components in (99). As in the RIP experiments, we pair each of 100 independent operators with 200 independently generated tensors and report the mean, over operators, of the maximum relative error on each test set. The measurement operator and its adjoint are implemented via the fast Walsh–Hadamard transform, and $\mathcal{P}_{\mathcal{V}_s}$ is evaluated as a coordinate mask.

In each experiment, we vary one parameter while holding the others fixed: the measurement budget $m \in \{150, 175, \dots, 800\}$ with $(d, s, k, p) = (10, 3, 3, 2)$; the tensor order $d \in \{9, \dots, 14\}$ with $(s, k, p, m) = (3, 3, 2, 500)$; and the interaction order $s \in \{1, \dots, 7\}$ with $(d, k, p, m) = (12, 4, 1, 800)$. Here, p denotes the step size between consecutive sliding windows.

To compare the observed trends with Theorem 5.2 and Lemma 5.3, we use the scaling guides

$$g_{\text{bp}}(d, s, m) := \left[\frac{D_s}{m} \log \left(\frac{2D_s}{\delta} \right) \right]^{1/2}, \quad (102)$$

$$g_{\text{init}}(d, s, m) := (1 + \sqrt{d-1})g_{\text{bp}}(d, s, m),$$

with $\delta = 0.05$, for the projected backprojection and post-TT-SVD errors, respectively. When only m varies, both guides reduce to the $m^{-1/2}$ rate. Each guide is normalized independently to the first point of its empirical curve, so the comparison concerns scaling rather than the absolute constants in the bounds.

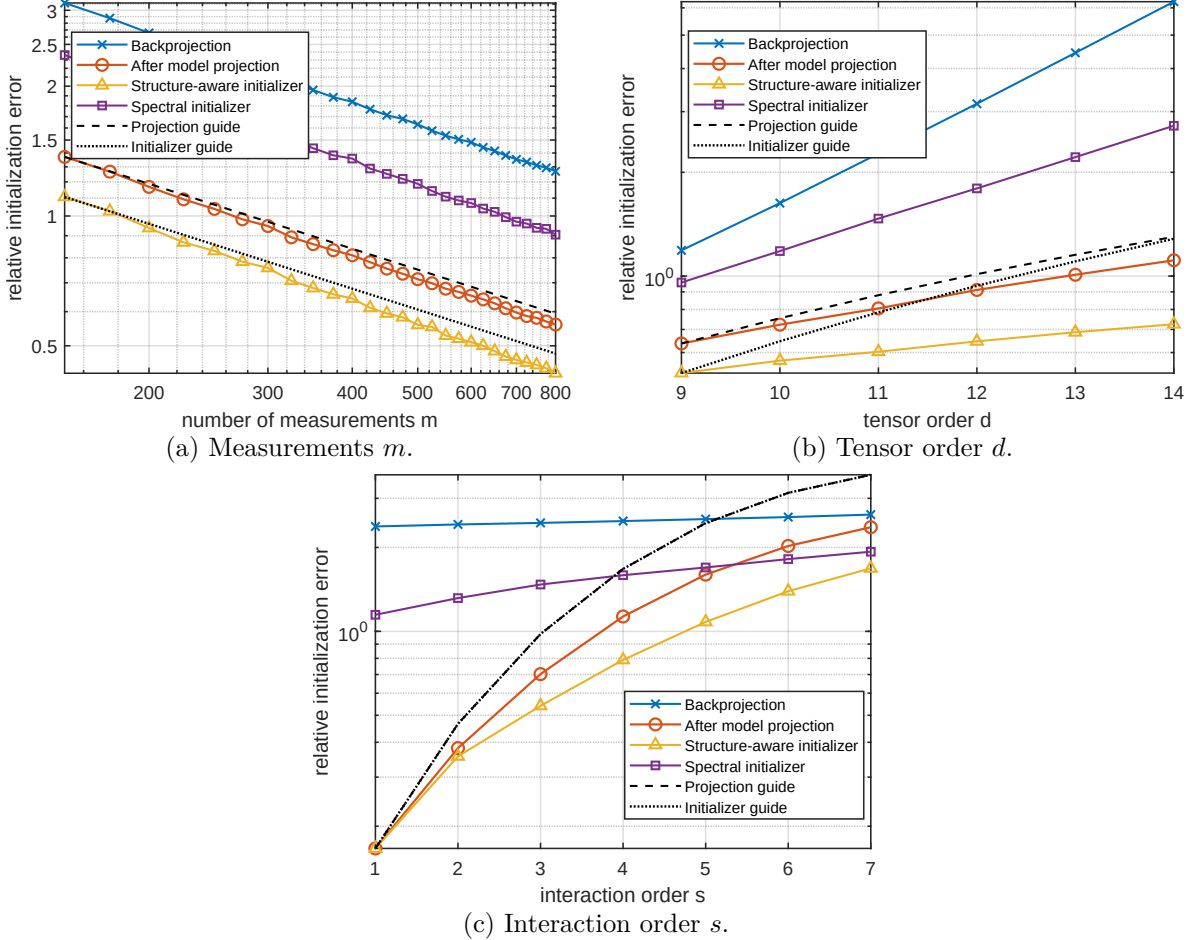


Figure 2: Empirical initialization errors at the three successive stages and for the spectral benchmark.

Figure 2 supports three conclusions. First, the projected backprojection and the post-TT-SVD initializer are consistent with the scaling predicted by Theorems 5.2 and 5.4, respectively. Second, the consistent ordering

$$e(\mathcal{A}^* \mathbf{y}) > e(\mathcal{P}_{\mathcal{V}_s} \mathcal{A}^* \mathbf{y}) > e(\text{TT-SVD}_{\mathbf{r}}(\mathcal{P}_{\mathcal{V}_s} \mathcal{A}^* \mathbf{y}))$$

shows that both the interaction-space projection and the TT-SVD truncation reduce the error, supporting the design of the proposed initializer. For the TT-SVD step, however, Lemma 5.3 guarantees only that the relative post-TT-SVD error is bounded by $(1 + \sqrt{d-1})e(\mathcal{P}_{\mathcal{V}_s} \mathcal{A}^* \mathbf{y})$.

In these experiments, TT-SVD therefore acts as a denoising step. Analyzing the spectra of the TT unfoldings of $\mathcal{P}_{\mathcal{V}_s} \mathcal{A}^* \mathbf{y}$ may lead to a sharper post-TT-SVD error bound. Third, the structure-aware initializer has lower error than $\text{TT-SVD}_r(\mathcal{A}^* \mathbf{y})$ throughout, showing that the additional model projection improves on the standard spectral initializer in this setting.

6.3 Initialization in the full ALS recovery scheme

The preceding experiments assess the initializer before optimization; we now test it within the full fixed-rank ALS scheme of Section 3.1. Within each paired trial, the methods differ only in initialization: they use the same ground-truth tensor, Walsh–Hadamard measurements, oracle TT ranks, and fixed-rank ALS recovery algorithm. The proposed structure-aware initializer is $\text{TT-SVD}_r(\mathcal{P}_{\mathcal{V}_s} \mathcal{A}^* \mathbf{y})$. We compare it with the standard unprojected spectral initializer $\text{TT-SVD}_r(\mathcal{A}^* \mathbf{y})$, which isolates the effect of the low-order projection.

The two uninformative baselines are a random TT tensor generated by `tt_random` with the oracle ranks and a near-zero initializer. Because an exact zero tensor has no nondegenerate prescribed-rank TT-SVD factors, the latter is formed by scaling a Gaussian vector to $10^{-6} \|\mathcal{X}^*\|_F$ and truncating it to the oracle ranks. For each parameter value, we run 100 independent tensor–operator trials, each for at most 100 ALS sweeps. Success means a final relative error below 10^{-5} . We report the success rate, mean final relative error, and mean first-passage sweep count, assigning a value of 101 to trials that do not reach the tolerance within 100 sweeps.

Varying the tensor order. We first vary the tensor order from $d = 9$ to 16, while fixing the interaction order at $s = 3$, the number of active components at $k = 3$, the window step size at $p = 2$, and the measurement budget at $m = 400$. Thus, the ambient dimension grows from 2^9 to 2^{16} while the local interaction structure and measurement budget remain fixed.

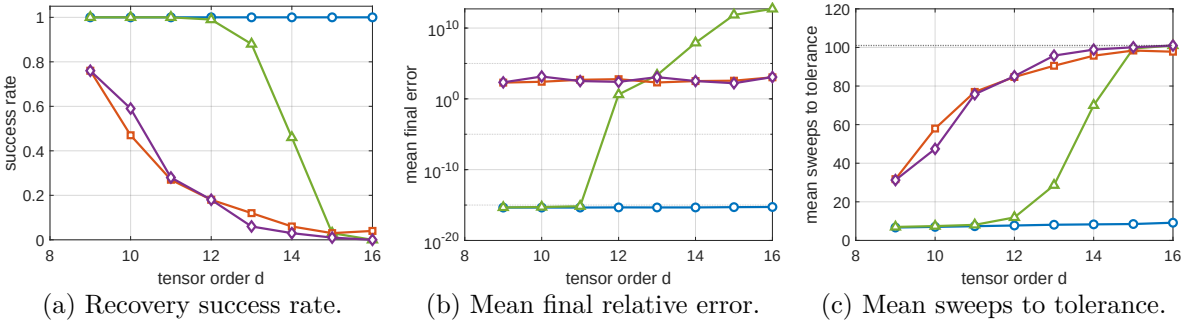


Figure 3: Full ALS recovery versus tensor order for $m = 400$, based on 100 trials. Initializers: structure-aware (blue circles), spectral (green triangles), random TT (orange squares), and near-zero (purple diamonds).

At fixed $m = 400$, the structure-aware initializer recovers every trial at every tested tensor order, with near-machine-precision final error and only a modest increase in sweep count. In contrast, the spectral initializer becomes unreliable as d grows and fails all trials at $d = 16$; the uninformative baselines deteriorate as well. Since the two spectral-type methods differ only by $\mathcal{P}_{\mathcal{V}_s}$, their separation isolates the benefit of removing off-model components before TT-SVD.

Varying the number of measurements. We next vary the measurement budget as $m = 100, 150, \dots, 1200$, while fixing the tensor order at $d = 12$, the interaction order at $s = 3$, the

number of active components at $k = 3$, and the window step size at $p = 2$.

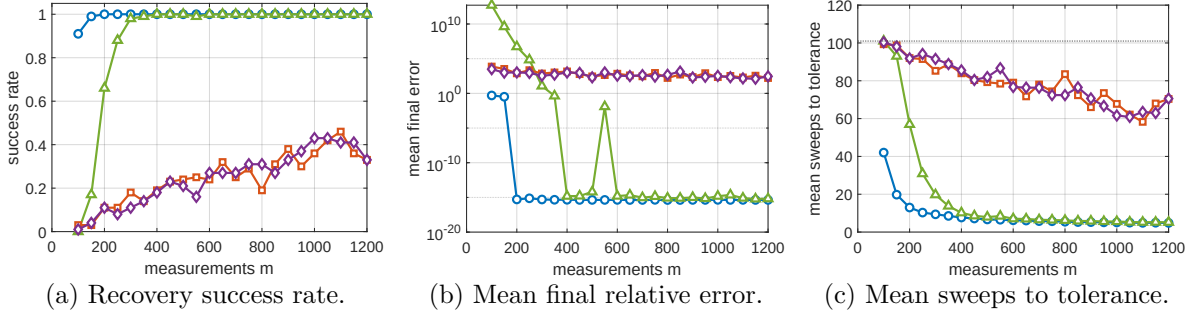


Figure 4: Full ALS recovery versus the number of measurements for $d = 12$, based on 100 trials. Initializers: structure-aware (blue circles), spectral (green triangles), random TT (orange squares), and near-zero (purple diamonds).

The measurement sweep shows that the projection is most valuable in the undersampled regime. The structure-aware method succeeds in every trial from $m = 200$ onward, whereas the spectral initializer becomes comparably reliable only around $m = 300$ – 400 ; the uninformative baselines remain unreliable throughout. The isolated spikes in the mean final error of the spectral initializer are caused by occasional failed trials: successful trials reach near-machine precision, whereas a failed trial can retain a relative error close to one and therefore dominate the mean. Once both spectral-type methods converge, they achieve similar final accuracy with similar sweep counts. Thus, the principal benefit of the structure-aware initializer is a higher probability of convergence with fewer measurements, rather than greater accuracy after convergence.

These conclusions are limited to the noiseless sliding-window model with fixed supports and oracle TT ranks.

7 Conclusion

This work develops a compressed-sensing framework for length- 2^d signals whose binary tensorizations combine bounded TT rank with sparse low-order interaction structure. For randomly subsampled Walsh–Hadamard measurements, we established a uniform RIP on this model class and used it to prove uniqueness of noiseless recovery. We also introduced a structure-aware initializer for fixed-rank ALS and proved a uniform initialization error bound with a measurement requirement $\mathcal{O}(d^{s+1} \log d)$ for fixed interaction order s , target accuracy, and failure probability. The experiments support the predicted qualitative scaling, while showing milder dependence on d , s , and k than the worst-case bounds. Most importantly, the structure-aware initializer yields lower initialization error and reliable ALS recovery with fewer measurements than the standard spectral initializer.

The analysis suggests two concrete directions for sharper theory. For the uniform RIP, a single-scale covering argument already gives a measurement bound of comparable polylogarithmic order to general sparse RIP results. The milder empirical dependence on d , s , and k suggests that chaining or a tighter model-specific covering argument could sharpen the remaining factors. For the initializer, the current proof controls the projected backprojection over the full interaction space $\mathcal{V}_s$ and then applies the generic TT-SVD factor $1 + \sqrt{d-1}$. In these experiments, TT-SVD acts as a denoising step. Analyzing the spectra of the TT unfoldings of $\mathcal{P}_{\mathcal{V}_s}(\mathcal{A}^* \mathbf{y})$ may lead to a sharper post-TT-SVD error bound.

The concentration argument also suggests extensions to uniformly subsampled orthonormal systems with bounded entries. One example is the one-dimensional discrete Fourier transform applied to the ground-truth vector $\mathbf{x}^*$, with its Fourier coefficients sampled uniformly. The present QTT formulation reshapes a finely discretized one-dimensional signal of length 2^d into a d -mode tensor with binary modes [15, 29]. Separately, tensors with binary modes arise intrinsically in quantum many-body models, where low-rank matrix product representations and few-body interactions suggest a natural, though not automatic, connection to our joint structural model [28, 13]. This suggests a possible direction for structure-aware recovery in physical inverse problems, but the present evidence is limited to noiseless, exactly structured sliding-window tensors with fixed supports and oracle TT ranks. Realizing this potential will require verifying the simultaneous presence of low TT rank and sparse low-order interactions, developing physically realizable measurement schemes, and addressing noise, model mismatch, and unknown ranks.

Acknowledgments

This work grew out of the author’s master’s thesis at the University of Chicago. The author thanks his supervisor, Yuehaw Khoo, for his guidance and many helpful discussions. The author used ChatGPT to assist with language editing and stylistic refinement of the manuscript. All mathematical content was independently reviewed and verified by the author.

Statements and Declarations

Funding. No funding was received for conducting this study.

Competing interests. The author has no relevant financial or non-financial interests to disclose.

Data availability. All data analyzed in this study were generated synthetically. The generated data supporting the findings are included in the accompanying code repository at https://github.com/Jingchun-Shao/compressed_sensing_with_QTT.

Code availability. The MATLAB code used to generate the data and figures and to perform the reported recovery experiments is available at https://github.com/Jingchun-Shao/compressed_sensing_with_QTT.

Author contributions. The author developed the methodology and software, performed the analysis and experiments, and wrote and approved the manuscript.

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